

Literature Review: Optimizer Comparison in Neural Network Training

Literature Review — Optimizer Comparison Papers

Three papers on neural network optimizer comparison were reviewed for this task.

Paper 1 — Soydaner (2020), “A Comparison of Optimization Algorithms for Deep Learning”

- **Objective:** Systematically benchmark widely-used optimizers — SGD variants, AdaGrad, AdaDelta, RMSProp, Adam, AdaMax, Nadam, AMSGrad — across both supervised classification and unsupervised autoencoder reconstruction tasks, as network depth and dataset difficulty increase.
- **Datasets:** MNIST, CIFAR-10, Kaggle Flowers, and Labeled Faces in the Wild (LFW), using three CNN depths for classification and two autoencoder depths for reconstruction.
- **Optimizers:** SGD, SGD+Momentum, SGD+Nesterov, AdaGrad, AdaDelta, RMSProp, Adam, AdaMax, Nadam, AMSGrad.
- **Key findings:** Adam-family methods (Adam, AdaMax, Nadam, AMSGrad) were the most consistently strong performers overall. Plain SGD’s accuracy fell off sharply on the more difficult LFW dataset, and AdaGrad lagged throughout. AdaDelta stood out for needing essentially no learning-rate tuning while staying competitive.
- **Limitations:** Single-GPU experiments with hyperparameters mostly inherited from prior papers rather than tuned per optimizer; scope limited to vision tasks only.

Analysis

The widening gap between SGD and adaptive optimizers as task difficulty increases is a useful signal for choosing an optimizer under a tight tuning budget — though the exact ranking is dataset- and architecture-dependent, not fixed.

Paper 2 — Almog (2025), “An Analysis of Optimizer Choice on Energy Efficiency and Performance in Neural Network Training”

- **Objective:** Measure the three-way trade-off between accuracy, training time, and carbon emissions across optimizer choices — motivated by the environmental cost of ML training.
- **Datasets:** MNIST, CIFAR-10, CIFAR-100, with 8 optimizers $\times$ 15 random seeds (360 runs total) on Apple M1 Pro hardware, using CodeCarbon for emissions tracking.
- **Optimizers:** SGD, Adam, AdamW, RMSprop, Adagrad, Adadelata, Adamax, NAdam.
- **Key findings:** AdamW and NAdam were the most consistently efficient optimizers across datasets. SGD achieved the best accuracy specifically on the hardest dataset (CIFAR-100) but at a substantially higher emissions cost. Training duration and emissions were only weakly correlated — a fast optimizer isn’t automatically a low-emissions one.
- **Limitations:** Single hardware platform (Apple M1 Pro, not representative of GPU clusters); hyperparameters fixed from prior literature; only small CNNs on image-classification benchmarks were tested.

Analysis

Adds a sustainability dimension largely absent from Papers 1 and 3 — the “best” optimizer depends on whether accuracy, speed, or environmental cost is the priority, and these can point to different choices for the same dataset.

Paper 3 — “Comparative Study of Deep Learning Optimizers: SGD, Momentum SGD, RMSProp, AMSGrad, Adam, Yogi, and Lion” (2025, IJISRT)

- **Objective:** Compare seven optimizers, including the newer Yogi and Lion methods, on training/test accuracy, training loss, and sensitivity to learning-rate choice.
- **Datasets:** MNIST and CIFAR-10.
- **Optimizers:** SGD, Momentum SGD, RMSProp, AMSGrad, Adam, Yogi, Lion.
- **Key findings:** A high-learning-rate SGD ($\eta = 0.5$) achieved the best test accuracy on MNIST (99.14%) and the best training accuracy (99.89%); Momentum SGD and Adam were close behind (99.15% and 98% test accuracy) at a much lower learning rate ($1e-2$). The paper’s related-work summary also cites Adam reaching top test accuracy in other studies (0.9826 on MNIST, 0.9855 on CIFAR-10), with RMSProp excelling specifically on training accuracy in those comparisons.
- **Limitations:** Only two relatively easy benchmark datasets were used; the newer optimizers (Yogi, Lion) underperformed the established methods in this particular study, which may not generalize to larger or more complex tasks.

Analysis

Reinforces a pattern seen in Paper 1 — a carefully tuned SGD can match or exceed adaptive optimizers on simpler benchmarks, but adaptive methods converge faster and are far more forgiving of learning-rate choice, which matters more as problems scale up.

Comparison at a Glance

Paper	Datasets	Optimizers Tested	Primary Focus
Soydaner (2020)	MNIST, CIFAR-10, Kaggle Flowers, LFW	10 (SGD variants through AMSGrad)	Accuracy across increasing task difficulty
Almog (2025)	MNIST, CIFAR-10, CIFAR-100	8 (SGD through NAdam)	Accuracy vs. speed vs. energy/carbon cost
IJISRT (2025)	MNIST, CIFAR-10	7 (SGD through Yogi, Lion)	Accuracy and learning-rate sensitivity

Table 1: Summary of the three reviewed optimizer-comparison papers.

Conclusion

Across all three papers, a consistent, practical takeaway emerges: **Adam-family optimizers are the safest default when tuning time is limited**, and this advantage grows as datasets/models get more complex (Papers 1 and 2). **Well-tuned SGD remains competitive, sometimes even superior, on simpler problems** (Papers 1 and 3), but demands more careful learning-rate selection and, per Paper 2, can cost more in energy on harder tasks. No optimizer wins on every axis — accuracy, convergence speed, tuning effort, and energy cost — at once.

Taken together, the three papers also suggest an open gap worth flagging for future work: none of them jointly optimize for *all four* axes — accuracy, convergence speed, tuning effort, and energy cost — on the same hardware and dataset. A follow-up study benchmarking Adam-family and newer optimizers (Yogi, Lion) under a fixed energy budget, rather than a fixed epoch count, would give a more actionable picture for resource-constrained training scenarios such as edge or embedded deployment.

References

- [1] Soydaner, D. (2020). *A Comparison of Optimization Algorithms for Deep Learning*. International Journal of Pattern Recognition and Artificial Intelligence, 34(13).
- [2] Almog, D. (2025). *An Analysis of Optimizer Choice on Energy Efficiency and Performance in Neural Network Training*. Preprint.
- [3] IJISRT (2025). *Comparative Study of Deep Learning Optimizers: SGD, Momentum SGD, RMSProp, AMSGrad, Adam, Yogi, and Lion*. International Journal of Innovative Science and Research Technology.